CASE STUDY

Chorismate mutase (2): nudged elastic band

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from runs with the 3.2.0 development version.

The scan of part 1 imposed a coordinate, d(C5–O7) − d(C1–C9). The nudged elastic band (NEB) needs only the reactant and the product: a chain of images between them is optimized until it follows the minimum energy path. Here it checks whether the scan coordinate missed a lower path, and it shows two practical points about NEB in enzymes: the choice of the end points and the number of images.

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1. The end points

Reactant: the starting structure of the system (`0_coords`). **Product**: the last structure of the scan of part 1 (C1–C9 = 1.56 Å, C5–O7 = 2.99 Å). Import the scan folder `CM_scan.ptGeo` with **File → Import Data...** (*pkl folder*, with its `output.log`) as the object `CM_scan` .

An object with several frames gives the NEB its **last** frame: choosing `CM_scan` as the product takes the last structure of the scan.

Why not optimize a product separately? We first made the product as in the original version of this tutorial: a harmonic restraint of 1.5 Å on C1–C9, an optimization, the restraint removed (**System → Remove → Restraints**) and a second optimization. That product is a fine minimum, but it differs from the reactant in the positions of many water molecules and side chains that moved during the optimizations. The NEB had to follow all those changes as well: after 600 iterations the chain had not converged and had three maxima. The structures of a scan change continuously, the environment included; taking both ends from it (or relaxing one from the other) gives the NEB only the chemistry to follow.

2. The NEB

Simulate → Nudged Elastic Band (NEB):

Field	Value
Reactants / Products	`0_coords` / `CM_scan` (its last frame)
Fixed Terminal Images	on
Number of Structures	**31**
Maximum Iterations / RMS Grad. Tolerance	1000 / 0.1
Spring Force Constant	500
Use Spline Redistribution	on (default)
Trajectory Name	`CM_NEB`

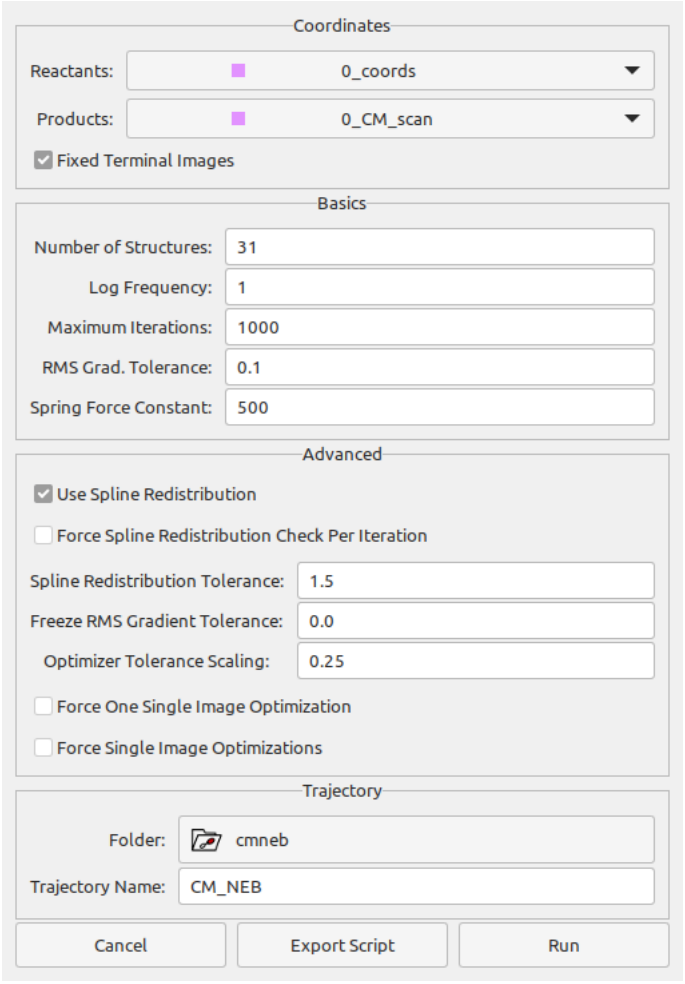


Figure 1. The NEB of chorismate mutase.

The chain converged in 191 iterations (5 minutes on one core). Import it from the *Finished!* dialog and open **Analysis → PES Analysis**.

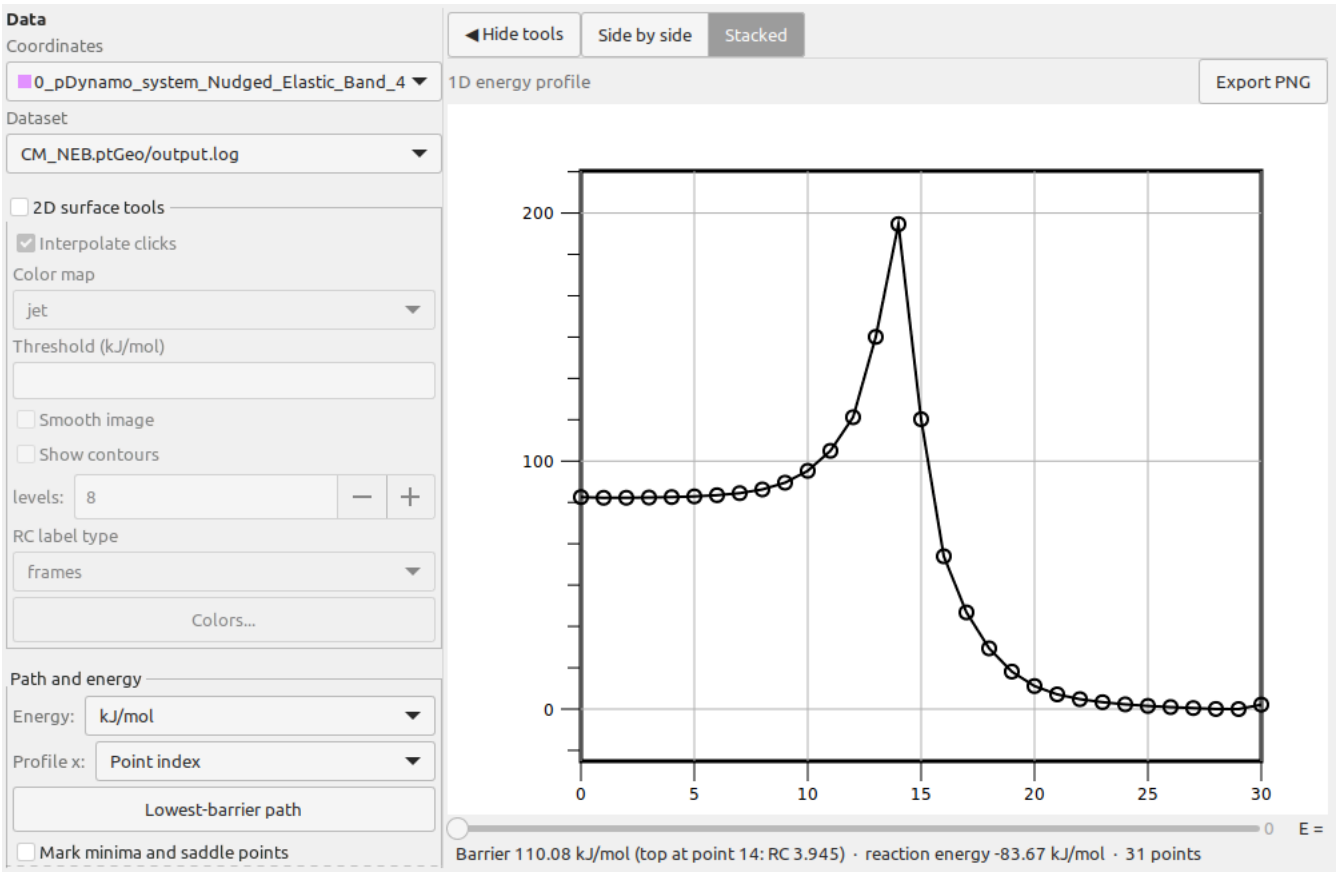


Figure 2. Energy along the 31 images (relative to the lowest image).

	Image	C5–O7 (Å)	C1–C9 (Å)	E − E(reactant) (kJ/mol)
Chorismate	0	1.48	3.15	0.0
Highest image	14	1.81	2.03	**110.1**
Prephenate	29	3.0	1.56	−85.5

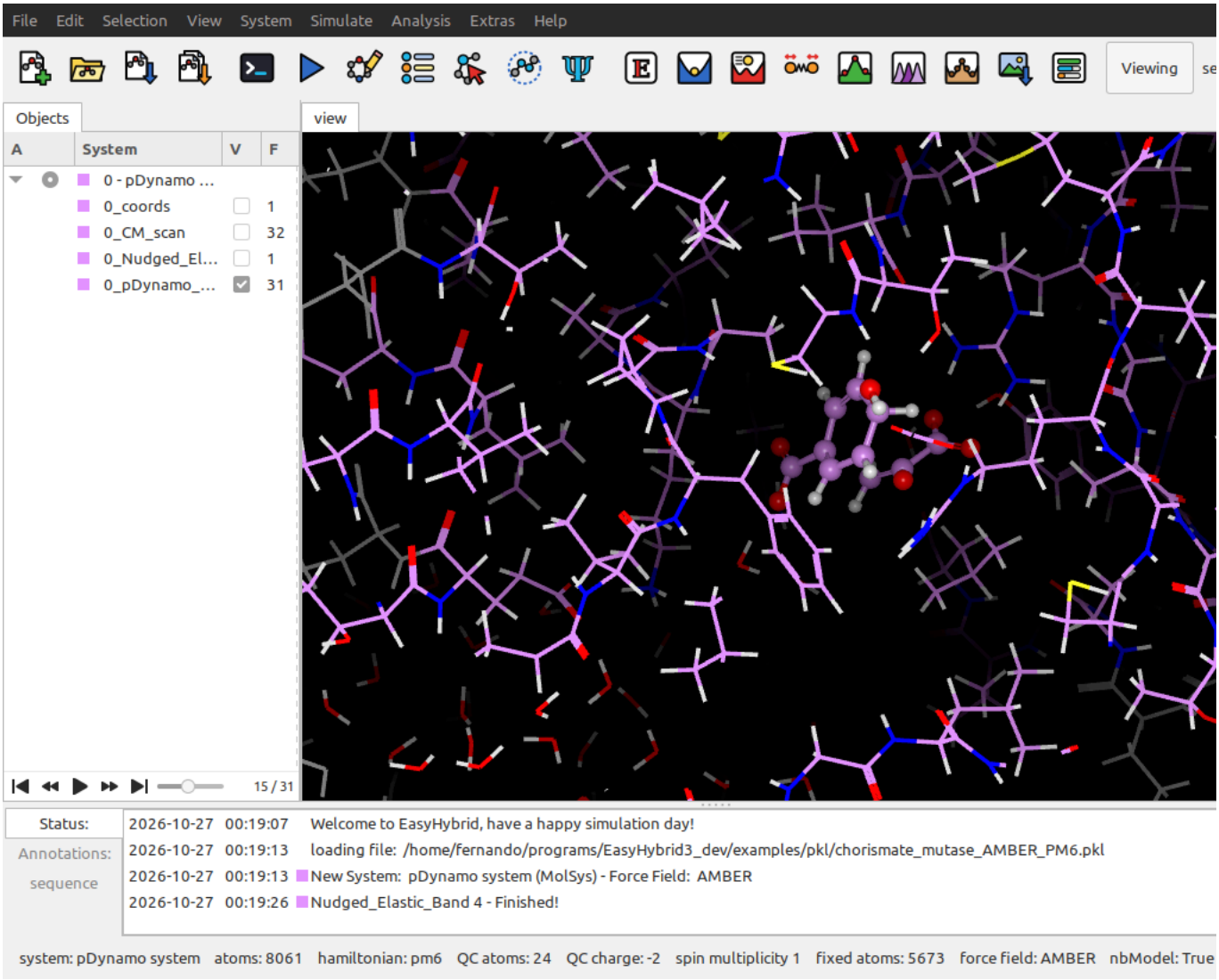


Figure 3. Image 14: C5–O7 breaking ahead of C1–C9 forming.

The NEB finds the same transition region as the scan (C5–O7 = 1.77, C1–C9 = 2.05 Å in the scan) and the same barrier within 2 kJ/mol: the coordinate of part 1 was a good one.

3. How many images?

The images are spread evenly along the whole path, and in an enzyme most of the path length is the motion of the environment, not of the bonds that change. With **15 structures** this NEB converged too, but the bond change fell between two images (C5–O7 = 1.56 Å in one, 2.00 Å in the next): the highest image was 56 kJ/mol above the reactant, a barrier underestimated by half. With 31 images the top is resolved. Look at the reacting distances of the images next to the highest one: if they jump, add images (or refine the top with **Simulate → Search Transition State (Baker)**).

Notes

- **Spring force constant.** A much stiffer chain (3000 here) made the optimization unstable; keep the default unless the images bunch together.
- **Potential energies.** Like the scan, the NEB path is a single relaxed structure per image. Part 3 computes the free energy with umbrella sampling.
- Reference: G. Mills, H. Jónsson, *Phys. Rev. Lett.* **1994**, 72, 1124.